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Satheesh Meadi

Graduate Research Assistant
Irrigation Research Laboratory
Biosystems & Agricultural Engineering
Oklahoma State University
202 Agricultural Hall, Stillwater, OK 74078
+1 (667) 433-1382 | satheesh.meadi@okstate.edu

Summary

Research-focused professional with a strong foundation in computer science, data science, and machine learning, with a particular interest in applying artificial intelligence to agriculture and environmental systems. Passionate about building data-driven models and intelligent solutions that improve sustainability, optimize resource use, and support better decision-making in modern farming.

Education

Oklahoma State University — Stillwater, Oklahoma
PhD in Biosystems and Agricultural Engineering — *Jan 2026 – Present*

University of Maryland, Baltimore County — Baltimore, Maryland
Master of Science in Data Science, CGPA: 3.97/4.0 — *Aug 2023 – May 2025*

Mohan Babu University (formerly Sree Vidyanikethan Engineering College) — Tirupati, India
Bachelor of Technology in Computer Science, CGPA: 3.60/4.0 — *Aug 2019 – Apr 2023*

Professional Positions Held

Graduate Research Assistant — Stillwater, Oklahoma
Biosystems & Agricultural Engineering, Oklahoma State University (OSU) — *Jan 2026 – Present*

- Extracted evapotranspiration (ET) raster data from Google Earth Engine and integrated ET, NDVI, precipitation, and reference evapotranspiration (ETo) variables to estimate irrigation requirements across five U.S. states.
- Built OLS, Random Forest, and LightGBM models to estimate height-adjusted crop canopy temperature from static sensors, ground sensors, and Mesonet weather variables.
- Reduced early-season temperature measurement error by 53% and achieved an RMSE of $1.52\,^{\circ}\mathrm{C}$, lowering manual labor and operational costs in agricultural thermal sensing.
- Investigating the effects of relative-humidity constraints on canopy temperature to improve irrigation scheduling across the 17-state U.S. Cotton Belt.
- Developing a centralized cotton canopy-temperature database and conducting field campaigns for sensor installation and data collection.

AI Research Fellow, Handshake Move Fellowship — Baltimore, Maryland
Handshake AI Solutions — *Aug 2025 – Dec 2025*
Project: CANARY – Domain-Specific Foundation Model Research

- Developed domain-adapted large language models for computer science, data science, and agricultural AI in collaboration with interdisciplinary research fellows.
- Fine-tuned transformer models on specialized datasets to improve domain intelligence, contextual reasoning, and factual precision.

Project: ORION – Scientific Integrity and Model Behavior Evaluation

- Conducted AI integrity evaluations to assess LLM reliability and behavior across complex machine learning workflows.
- Analyzed reasoning patterns and benchmarked transparency and decision consistency to support accountable AI system design.

Graduate Assistant Baltimore, Maryland
University of Maryland, Baltimore County (UMBC) *Aug 2024 – Apr 2025*

- Supported research and instruction for a Natural Language Processing and Large Language Models course focused on transformer architectures and generative AI.
- Evaluated student projects and research papers, providing feedback on model design, training efficiency, coherence, perplexity, and inference latency.
- Contributed to faculty research on improving transformer efficiency and coherence through reproducible experimentation.

Math Tutor Baltimore, Maryland
University of Maryland, Baltimore County (UMBC) *Jan 2024 – Aug 2024*

- Tutored 30+ students in advanced mathematics, strengthening analytical thinking and problem-solving skills relevant to data-driven environments.
- Applied performance tracking and collaborative support strategies that contributed to a 20% increase in test scores.

Graduate Engineer Intern Bangalore, India
LTI Mindtree *Jan 2023 – Aug 2023*

- Improved data ingestion throughput by 25% by engineering Python- and SQL-based ETL pipelines that consolidated JSON and CSV datasets into PostgreSQL marts.
- Reduced data quality errors by 30% through automated validation and governance checks embedded in the ETL workflow.
- Enhanced observability and SLA reliability by integrating Airflow DAG monitoring with AWS CloudWatch alerting.

Research & Publications

Canopy Temperature Sensor Data Optimization for Irrigation Scheduling *Paper*

- Developed and evaluated machine learning models to estimate height-adjusted crop canopy temperature using static sensors, ground sensors, irrigation data, and weather variables.
- Demonstrated that an Ordinary Least Squares (OLS) model reduced early-season measurement error by 53%, providing a cost-effective and automated solution for in-situ thermal sensing in agricultural water

management.

Extreme Learning Mechanism for Classification and Prediction of Soil Fertility Index *Paper*

- Applied machine learning techniques to classify soil nutrient indices and predict pH levels for the southeast laterite region of Andhra Pradesh, improving precision in soil fertility assessment.
- Leveraged agricultural soil data and feature-based modeling to support fertilizer recommendations and sustainable crop management aligned with India's Soil Health Program.

Stock Price Prediction by Normalizing LSTM and GRU Models *Paper*

- Developed a hybrid deep learning framework combining LSTM and GRU architectures to predict next-day stock prices in volatile, nonlinear markets.
- Designed and optimized a forecasting pipeline to support more reliable and data-driven financial analysis.

Peer-Reviewed Journal Publications & Conference Proceedings

1. **Meadi, S.**, Kalayambakam Janardhan, A. Y., Datta, S., & Das, H. (2026). Canopy Temperature Sensor Data Optimization for Irrigation Scheduling. 2026 IEEE DCAS Conference, Dallas, TX, April 10–12, 2026. *DOI:* 10.1109/DCAS69364.2026.11545108
2. Ganesh, D., Abinava Yeshwanth, K. J., **Meadi, S.**, Guru Sai Vignesh Reddy, M., Chirudeep, T., & Naveen Kumar Polisetty, S. (2022). Extreme Learning Mechanism for Classification & Prediction of Soil Fertility Index. *Journal of Pharmaceutical Negative Results, 13* (Special Issue 06), 37–43. *DOI:* 10.47750/pnr.2022.13.S06.006
3. Venkatarao, J., Chakravarthy, V. D., **Meadi, S.**, Durga, K. L., Pranay, K., & Purushotham, N. (2023). Stock Price Prediction by Normalizing LSTM and GRU Models. *Journal of Survey in Fisheries Sciences, 10*(1S), 5326–5332. *Paper:* ResearchGate

Conference Presentations – Oral

1. Datta, S., **Meadi, S.**, & Yeshwanth, A. (2026, July 12–15). Paving the Path Forward to a National Scale Irrigation Scheduler by Leveraging PyFAO56. 2026 ASABE Annual International Meeting (Agenda; Tool), Indianapolis, IN.
2. Datta, S., **Meadi, S.**, & Kundu, P. K. (2026, July 12–15). Data-Driven Modeling of Canopy Temperature for Variably Irrigated Cotton. 2026 ASABE Annual International Meeting (Agenda), Indianapolis, IN.
3. **Meadi, S.**, Datta, S., Kundu, P. K., Mahan, J., Taghvaeian, S., & Kennedy, T. (2026, July 12–15). Humidity Constraints on Canopy Temperature-Based Irrigation Scheduling Across Climatic Gradients in Cotton. 2026 ASABE Annual International Meeting (Agenda), Indianapolis, IN.
4. Pandit, A., Datta, S., **Meadi, S.**, & Taghvaeian, S. (2026, July 12–15). Seasonal Irrigation Estimation Framework Using OpenET. 2026 ASABE Annual International Meeting (Agenda), Indianapolis, IN.
5. **Meadi, S.**, Kalayambakam Janardhan, A. Y., & Datta, S. (2026, July 12–15). Can We Accurately Estimate Canopy Temperature Without Adjusting Sensor Heights to Reduce Labor Requirements? 2026 ASABE Annual International Meeting (Agenda), Indianapolis, IN.

6. **Meadi, S.**, Kalayambakam Janardhan, A. Y., Datta, S., & Das, H. (2026, April 10–12). Canopy Temperature Sensor Data Optimization for Irrigation Scheduling. 2026 IEEE DCAS Conference, Dallas, TX.

Academic Service

Ph.D. Admissions Interview Panel Member *2026*

- Served on an interview panel for prospective doctoral students applying to the Biosystems & Agricultural Engineering graduate program at Oklahoma State University in support of the Dr. Sumon Datta Irrigation Research Laboratory.

Skills

Languages	Python, Java, C++, SQL, Bash
Big Data & Processing	Apache Spark, PySpark, Hadoop, Kafka, Spark Streaming
Data Warehousing	Snowflake, Redshift, BigQuery, PostgreSQL, MySQL, Delta Lake
Cloud Platforms	AWS (S3, EC2, Glue, Lambda, Redshift), GCP (BigQuery, Dataflow), Azure (Data Factory, Synapse)
Data Governance & Observability	Schema Evolution, CloudWatch, Datadog
CI/CD & Infrastructure	GitHub Actions, Docker, Kubernetes, Terraform
Visualization	Tableau, Power BI, Streamlit, Plotly, Seaborn
ML/AI Tools	Hugging Face, MLflow, SageMaker, NLP (spaCy, NLTK), LLMs (RAG, GenAI)
Agricultural & Geospatial Analytics	Google Earth Engine (GEE), OpenET, Remote Sensing, Geospatial Analysis, Raster Data Processing, ET/ETo, NDVI, Precipitation Data, Irrigation Scheduling, Crop Canopy Temperature, Precision Agriculture
Field & Research Methods	Mesonet Weather Data, Relative Humidity Analysis, Field Sensor Installation, Sensor Data Collection, Canopy Temperature Data Management

Research & Technical Projects

Seasonal Irrigation Estimation Framework Using OpenET *Ongoing*

- Integrating OpenET evapotranspiration products with Google Earth Engine-derived NDVI, precipitation, and reference evapotranspiration (ETo) data to estimate seasonal irrigation requirements across multiple U.S. regions.

Humidity Constraints on Canopy Temperature-Based Irrigation Scheduling *Ongoing*

- Investigating how relative-humidity conditions affect canopy temperature and irrigation-scheduling decisions across climatic gradients in the U.S. Cotton Belt.

Canopy Temperature Estimation Without Sensor-Height Adjustment *2026*

- Evaluating whether crop canopy temperature can be estimated accurately without sensor-height adjustment to reduce field labor requirements while supporting irrigation decisions.

Canopy Temperature Sensor Data Optimization for Irrigation Scheduling *2026*

- Developed machine-learning models using static sensors, ground sensors, irrigation data, and weather variables; an OLS model reduced early-season measurement error by 53%.

Medi-Text Summarizer: Medical Text Analysis Application *Link available on request*

- Built an AI-driven summarization platform using PubMedBERT, DistilBERT, and LLaMA-3.2 to generate concise, context-aware summaries of medical research papers and clinical notes.
- Implemented named entity recognition pipelines and real-time Q&A features, improving summarization speed by 60% while maintaining 95% semantic accuracy on benchmark datasets.
- Deployed the application with Streamlit and Google Cloud Translation API support, enabling multilingual summarization for 200+ healthcare users across 15+ languages.

Dribble, Predict, Win: Real-Time NBA Game Forecasts *Link available on request*

- Developed a sports analytics system using historical player and team statistics from 10,000+ records to forecast NBA game outcomes with 92% accuracy.
- Built PySpark data pipelines for live and historical feeds, improving data processing efficiency by 50%, and added drift monitoring to reduce misclassification rates by 20%.

Financial Data Pipeline and Risk Analysis Dashboard *Link available on request*

- Engineered a financial data platform that ingested real-time market, stock, and portfolio data into PostgreSQL and Snowflake, improving ETL throughput by 40%.
- Implemented CAPM-based risk analysis and built dashboards for asset performance, risk ratios, and Sharpe metrics, reducing report turnaround time by 50%.
- Improved reliability through Airflow retry strategies and Kubernetes-based orchestration, lowering pipeline failures by 25%.